VANISHING OF THE HIGHER ETALE COHOMOLOGY OF THE PUSHFORWARD OF THE MULTIPLICATIVE GROUP FROM THE GENERIC POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the spectral sequence is written out and both vanishing inputs, including the degree bookkeeping, are identified explicitly.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The Leray spectral sequence for the inclusion of the generic point. Same chapter, the Leray spectral sequence of the morphism j : eta -> X for etale cohomology. Supplies the spectral sequence on which the argument runs.
- Vanishing of the cohomology of the multiplicative group on the generic point of a curve. Same chapter, Lemma lemma-cohomology-Gm-function-field-curve. Kills the target of the spectral sequence in degrees at least one.
- Vanishing of the higher direct images of the multiplicative group. Same chapter, Lemma lemma-higher-direct-jstar-Gm. Kills the terms with q at least one on the E_2 page.

Source reference. Original lemma 03RG: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10907–10907. Navigation pointer invoked at proof line 11017; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03RJ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10988–10988. Navigation pointer invoked at proof

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line 11048; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *For all $p \geq 1$, $H^p_{\acute{e}tale}(X, j_*\mathbf{G}_{m,\eta}) = 0$.*

Proof. The Leray spectral sequence reads

$$E_2^{p,q} = H^p_{\acute{e}tale}(X, R^q j_*\mathbf{G}_{m,\eta}) \Rightarrow H^{p+q}_{\acute{e}tale}(\eta, \mathbf{G}_{m,\eta}),$$

The right hand side vanishes for $p + q \geq 1$ by Lemma 03RG. The left hand side vanishes for $q \geq 1$ by Lemma 03RJ. Thus we get the desired vanishing. □